12TH CHEMISTRY · CHAPTER 32 · FULL CHAPTER MCQS

Medicine, Agriculture and Industry

141 topic-wise MCQs from the complete chapter with answers and reasons

32.1 THERAPEUTIC INDEX AND WINDOW

1. The science that deals with what a drug does and its safety margins is:

(a) Physiology (b) Pathology
(c) Toxicography (d) Pharmacology

Answer: (d) Pharmacology — Definition given in section 32.1.

2. The difference between a medicine and a poison lies mainly in the:

(a) Dose (b) Colour
(c) Taste (d) Brand

Answer: (a) Dose — Every drug may be a potential poison; the dose decides.

3. The range of plasma concentrations in which a drug is effective without harmful side effects is the:

(a) Duration of action
(b) Therapeutic index
(c) Therapeutic window
(d) Toxic dose

Answer: (c) Therapeutic window — It lies between MEC and MTC.

4. MEC stands for:

(a) Maximum effective concentration
(b) Minimum effective concentration
(c) Minimum excretion concentration
(d) Mean effective capacity

Answer: (b) Minimum effective concentration — Below it the drug has no clinical effect.

5. The lowest dose at which a drug begins to cause adverse effects is the:

(a) TDM (b) MEC
(c) ED_{50} (d) MTC

Answer: (d) MTC — Minimum toxic concentration.

6. The time for which the drug concentration stays within its therapeutic window is the:

(a) Duration of action
(b) Half-life
(c) Onset time
(d) Toxic period

Answer: (a) Duration of action — Definition in section 32.1.1.

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7. A wide therapeutic window means the drug is:

(a) More dangerous
(b) Safer, with flexible dosing
(c) Ineffective
(d) Always addictive

Answer: (b) Safer, with flexible dosing — Large gap between the working dose and the harmful dose.

8. For a drug with a narrow therapeutic window, doctors often:

(a) Stop all food
(b) Double the dose
(c) Check the patient's blood level
(d) Give it without limit

Answer: (c) Check the patient's blood level — The effective dose is very close to the toxic dose.

9. Which factor can affect the therapeutic window of a drug?

(a) Age and genetic factors
(b) Colour of the tablet
(c) Shape of the bottle
(d) Time of manufacture

Answer: (a) Age and genetic factors — Also obesity, health and use of several medicines.

10. The therapeutic index is a measure of a drug's:

(a) Colour (b) Safety
(c) Solubility (d) Price

Answer: (b) Safety — It compares the toxic dose with the effective dose.

11. TD_{50} is the dose that is toxic in:

(a) 0.5% of people (b) 5% of people
(c) 100% of people (d) 50% of people

Answer: (d) 50% of people — Toxic Dose 50.

12. ED_{50} is the dose that:

(a) Is toxic for 50% of people
(b) Kills 50% of people
(c) Works for 50% of people
(d) Works for all people

Answer: (c) Works for 50% of people — Effective Dose 50.

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13. A drug has ED_{50} = 10 mg and TD_{50} = 100 mg. Its TI is:

(a) 10
(b) 0.1
(c) 90
(d) 1000

Answer: (a) 10 — TI = 100 ÷ 10.

14. Which drug is the safest?

(a) One with TI = 2
(b) One with TI = 50
(c) One with TI = 5
(d) One with TI = 10

Answer: (b) One with TI = 50 — A higher TI means a larger safety margin.

15. Which of these drugs has a high therapeutic index?

(a) Both warfarin and digoxin
(b) Warfarin
(c) Digoxin
(d) Paracetamol

Answer: (d) Paracetamol — Penicillin and paracetamol have high TI.

16. Warfarin is a:

(a) Antibiotic
(b) Painkiller
(c) Blood thinner
(d) Antiviral

Answer: (c) Blood thinner — It has a low TI.

17. Digoxin is a medicine for the:

(a) Heart
(b) Skin
(c) Eyes
(d) Bones

Answer: (a) Heart — A low-TI heart medicine.

18. TDM stands for:

(a) Therapeutic dose maximum
(b) Toxic dose measurement
(c) Total drug mass
(d) Therapeutic drug monitoring

Answer: (d) Therapeutic drug monitoring — Regular blood tests for low-TI drugs.

19. Yellow-tinted vision caused by digoxin toxicity is called:

(a) Myopia
(b) Xanthopsia
(c) Euphoria
(d) Fibrillation

Answer: (b) Xanthopsia — Mentioned in the Did You Know box.

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32.2 IMPORTANCE OF pH IN STOMACH

20. The pH of the stomach is about:

(a) 8.0–9.0
(b) 5.5–6.5
(c) 7.0–7.4
(d) 1.5–3.5

Answer: (d) 1.5–3.5 — A highly acidic environment.

21. The acidity of the stomach is regulated by:

(a) Parietal cells
(b) Red blood cells
(c) Nerve cells
(d) Bone cells

Answer: (a) Parietal cells — They pump H^+ ions into the stomach.

22. The stomach protects itself from its acid by:

(a) Calcium ions
(b) A layer of fat
(c) Mucus and bicarbonate ions
(d) Chloride ions

Answer: (c) Mucus and bicarbonate ions — They neutralise acid at the cell surface.

23. One function of stomach acid is to:

(a) Make bile
(b) Kill bacteria
(c) Absorb water
(d) Make insulin

Answer: (b) Kill bacteria — It also helps digestion.

24. The chemical change of drugs by stomach acid is an example of:

(a) Mutation
(b) Opioid binding
(c) Non-specific reactions
(d) Polymerisation

Answer: (c) Non-specific reactions — The low pH can break down some drugs.

25. Some medicines have a protective coating so that they dissolve in the:

(a) Oesophagus
(b) Mouth
(c) Stomach
(d) Small intestine

Answer: (d) Small intestine — It is less acidic than the stomach.

26. Magnesium hydroxide is used as an:

(a) Antacid
(b) Antibiotic
(c) Antiviral
(d) Opiate

Answer: (a) Antacid — It neutralises excess HCl.

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32.3 MECHANISM OF DRUG ACTION

27. Aspirin acts by blocking the:

(a) Transpeptidase enzyme (b) COX enzyme
(c) Protease enzyme (d) Opioid receptor

Answer: (b) COX enzyme — Discussed in Chemistry XI.

28. Penicillin is an:

(a) Antacid (b) Antiviral
(c) Antibiotic (d) Opiate

Answer: (c) Antibiotic — It treats bacterial infections.

29. Which infection is treated with penicillin?

(a) Hepatitis C (b) Influenza
(c) HIV (d) Strep throat

Answer: (d) Strep throat — It is a bacterial infection.

30. Penicillin contains a beta-lactam ring and a:

(a) Thiazolidine ring (b) Benzene ring only
(c) Pyridine ring (d) Furan ring

Answer: (a) Thiazolidine ring — Two core rings.

31. The beta-lactam ring has:

(a) Three members (b) Four members
(c) Five members (d) Six members

Answer: (b) Four members — A four-membered ring.

32. The thiazolidine ring of penicillin has:

(a) Seven members (b) Four members
(c) Six members (d) Five members

Answer: (d) Five members — A five-membered ring containing S and N.

33. Bacterial enzymes that deactivate penicillin are called:

(a) Beta-lactamases (b) Proteases
(c) Lipases (d) Amylases

Answer: (a) Beta-lactamases — They attack the beta-lactam ring.

34. Penicillin kills bacteria by stopping the formation of a functional:

(a) Ribosome (b) Nucleus
(c) Cell wall (d) Flagellum

Answer: (c) Cell wall — The bacteria burst and die.

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35. Penicillin does not harm human cells because they have no:

(a) Mitochondria (b) Cell membrane
(c) Nucleus (d) Cell wall

Answer: (d) Cell wall — Its target is absent in human cells.

36. Penicillin was discovered by:

(a) Louis Pasteur (b) Alexander Fleming
(c) Fritz Haber (d) Robert Koch

Answer: (b) Alexander Fleming — In 1928.

37. Penicillin was discovered in:

(a) 1928 (b) 1828
(c) 1905 (d) 1953

Answer: (a) 1928 — From a contaminated petri dish.

38. The mould that produces penicillin is:

(a) Yeast (b) Rhizopus
(c) Penicillium (d) Mucor

Answer: (c) Penicillium — Identified after Fleming's observation.

39. Opiates come from the:

(a) Opium poppy plant
(b) Tobacco plant
(c) Tea plant
(d) Cotton plant

Answer: (a) Opium poppy plant — A class of powerful pain relievers.

40. Which of these is a legal opiate medicine?

(a) Heroin (b) Codeine
(c) Cocaine (d) Nicotine

Answer: (b) Codeine — Morphine, codeine, oxycodone and hydrocodone are legal drugs.

41. The body's own painkillers are called:

(a) Enzymes (b) Interferons
(c) Endorphins (d) Antibodies

Answer: (c) Endorphins — They bind to opioid receptors.

42. The number of types of opioid receptors is:

(a) Five (b) Two
(c) Four (d) Three

Answer: (d) Three — Mu, kappa and delta.

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43. The primary target of morphine is the:

(a) Mu receptor (b) Kappa receptor
(c) Delta receptor (d) COX enzyme

Answer: (a) Mu receptor — Mu receptors give analgesia.

44. Respiratory depression and euphoria follow activation of:

(a) Delta receptors (b) Mu receptors
(c) Kappa receptors (d) Parietal cells

Answer: (b) Mu receptors — A dangerous side effect of opioids.

45. Kappa receptors are involved in:

(a) Digestion of fat
(b) Mood regulation only
(c) Spinal analgesia and sedation
(d) Blood clotting

Answer: (c) Spinal analgesia and sedation — They are less associated with euphoria.

46. Delta receptors play a role in:

(a) Hearing
(b) Breathing only
(c) Vision
(d) Emotional response and mood

Answer: (d) Emotional response and mood — Stated in section 32.3.2.

47. When an opiate binds to its receptor it:

(a) Closes calcium and opens potassium channels
(b) Opens calcium and closes potassium channels
(c) Opens sodium channels only
(d) Destroys the neuron

Answer: (a) Closes calcium and opens potassium channels — The neuron is hyperpolarised.

48. Opiates cause addiction by excessive release of:

(a) Insulin (b) Dopamine
(c) Adrenaline (d) Thyroxine

Answer: (b) Dopamine — In the reward centre of the brain.

49. Codeine is used in cough syrups because it:

(a) Kills viruses
(b) Kills bacteria
(c) Neutralises acid
(d) Suppresses the cough reflex

Answer: (d) Suppresses the cough reflex — A use of opioids.

50. Heroin is chemically:

(a) Acetylsalicylic acid (b) Methylmorphine
(c) Diacetylmorphine (d) Dimethylcodeine

Answer: (c) Diacetylmorphine — It was once sold by Bayer.

51. Compared with morphine, the potency of heroin is about:

(a) Fifty times (b) Half
(c) Equal (d) Five times

Answer: (d) Five times — It crosses the blood–brain barrier faster.

32.4 TREATING VIRAL INFECTIONS

52. Viruses are best described as:

(a) Parasites (b) Independent organisms
(c) Bacteria (d) Fungi

Answer: (a) Parasites — They replicate only inside host cells.

53. Viruses replicate using:

(a) Sunlight
(b) Their own ribosomes
(c) The host's DNA/RNA machinery
(d) Cell walls

Answer: (c) The host's DNA/RNA machinery — This makes selective drugs hard to design.

54. A rapid change in the genetic material of a virus is called:

(a) Hydration (b) Mutation
(c) Fermentation (d) Leaching

Answer: (b) Mutation — It makes drugs and vaccines less effective.

55. Flu vaccines are updated every year because viruses:

(a) Become bacteria (b) Grow cell walls
(c) Die in winter (d) Mutate quickly

Answer: (d) Mutate quickly — The virus changes from year to year.

56. Present antiviral drugs can kill only:

(a) Dormant viruses (b) Active viruses
(c) All bacteria (d) Viruses in the genome

Answer: (b) Active viruses — Sleeping viruses escape them.

57. Protease inhibitors are used to treat:

(a) Cellulitis (b) Strep throat
(c) HIV and hepatitis C (d) Scarlet fever

Answer: (c) HIV and hepatitis C — They act by enzyme inhibition.

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58. Proteins that boost the immune system against viruses are:

(a) Interferons (b) Endorphins
(c) Opiates (d) Beta-lactamases

Answer: (a) Interferons — Some antivirals enhance immunity.

32.5 FERTILIZERS

59. Fertilizers are added to soil to:

(a) Kill weeds
(b) Kill insects
(c) Provide essential nutrients
(d) Reduce rainfall

Answer: (c) Provide essential nutrients — They improve fertility and yield.

60. Ammonia for synthetic fertilizers is made by the:

(a) Chlor-alkali process (b) Solvay process
(c) Contact process (d) Haber-Bosch process

Answer: (d) Haber-Bosch process — From N_2 and H_2.

61. Compost and animal manure are:

(a) Synthetic fertilizers (b) Organic fertilizers
(c) Pesticides (d) Herbicides

Answer: (b) Organic fertilizers — They come from natural sources.

62. The number of essential elements required by plants is:

(a) Seventeen (b) Seven
(c) Nine (d) Twenty-seven

Answer: (a) Seventeen — Micronutrients plus macronutrients.

63. Which is a micronutrient?

(a) Nitrogen (b) Zinc
(c) Calcium (d) Magnesium

Answer: (b) Zinc — Micronutrients: B, Fe, Cu, Cl, Mn, Mo, Zn.

64. Which is a macronutrient?

(a) Molybdenum (b) Boron
(c) Copper (d) Magnesium

Answer: (d) Magnesium — There are nine macronutrients.

65. The primary macronutrients are:

(a) N, P and K (b) C, H and O
(c) Ca, Mg and S (d) Fe, Cu and Zn

Answer: (a) N, P and K — Needed in the largest quantities.

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66. The most limiting factor in crop production is often:

(a) Boron
(b) Chlorine
(c) Nitrogen
(d) Zinc

Answer: (c) Nitrogen — It builds proteins and chlorophyll.

67. Phosphorus is a structural component of:

(a) Starch
(b) Chlorophyll
(c) Cellulose
(d) ATP

Answer: (d) ATP — ATP stores and releases energy.

68. Phosphorus is typically supplied as:

(a) Urea
(b) Superphosphates
(c) Potash
(d) Gypsum

Answer: (b) Superphosphates — Or ammonium phosphates.

69. Which nutrient acts as an osmoregulator?

(a) Potassium
(b) Nitrogen
(c) Phosphorus
(d) Carbon

Answer: (a) Potassium — It does not become part of the plant's structure.

70. Potassium activates more than:

(a) Six hundred enzymes
(b) Six enzymes
(c) Sixty enzymes
(d) Two enzymes

Answer: (c) Sixty enzymes — They facilitate plant metabolism.

71. Excess fertilizer washed into lakes causes:

(a) Overgrowth of algae
(b) More fish
(c) Clearer water
(d) Higher oxygen

Answer: (a) Overgrowth of algae — It harms aquatic life.

72. Crop rotation helps to:

(a) Increase pests
(b) Restore soil nutrients
(c) Lower soil fertility
(d) Cause acid rain

Answer: (b) Restore soil nutrients — It reduces the need for synthetic fertilizers.

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32.6 PESTICIDES

73. Pesticides are classified according to their:

(a) Smell (b) Colour
(c) Price (d) Target pest

Answer: (d) Target pest — Insecticides, herbicides and fungicides.

74. Organophosphates are:

(a) Insecticides (b) Herbicides
(c) Fungicides (d) Fertilizers

Answer: (a) Insecticides — They are neurotoxins for insects.

75. Many modern insecticides act as:

(a) Vitamins (b) Hormones
(c) Neurotoxins (d) Nutrients

Answer: (c) Neurotoxins — They cause paralysis and death.

76. Colony Collapse Disorder in honeybees is linked to:

(a) Triazoles (b) Neonicotinoids
(c) Superphosphates (d) Interferons

Answer: (b) Neonicotinoids — A risk to global pollination.

77. Chemicals that control weeds are:

(a) Antibiotics (b) Insecticides
(c) Fungicides (d) Herbicides

Answer: (d) Herbicides — Weeds are unwanted, competitive plants.

78. Some herbicides kill weeds by stopping:

(a) Rainfall (b) Pollination
(c) Photosynthesis (d) Nitrogen fixation in air

Answer: (c) Photosynthesis — They disrupt electron transport.

79. Fungicides target the synthesis of:

(a) Chlorophyll (b) Ergosterol
(c) Cellulose (d) Haemoglobin

Answer: (b) Ergosterol — It is needed for the fungal cell membrane.

80. Triazoles and strobilurins are examples of:

(a) Fungicides (b) Insecticides
(c) Herbicides (d) Fertilizers

Answer: (a) Fungicides — Named in section 32.6.

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81. When pests develop immunity to a pesticide, farmers are forced to use:

(a) No pesticide
(b) More toxic pesticides
(c) More water
(d) Less fertilizer

Answer: (b) More toxic pesticides — A risk of pesticide use.

32.7 ACID RAIN

82. Acid rain is a mixture of:

(a) Carbonic and citric acid
(b) Hydrochloric and ethanoic acid
(c) Nitric acid and sulphuric acid
(d) Phosphoric and boric acid

Answer: (c) Nitric acid and sulphuric acid — Formed from industrial pollution.

83. Normal rain water is slightly acidic because of:

(a) Carbonic acid
(b) Nitric acid
(c) Sulphuric acid
(d) Hydrochloric acid

Answer: (a) Carbonic acid — Dissolved CO_2; pH about 5.6.

84. The gases that cause acid rain are:

(a) CH_4 and Ar
(b) O_2 and N_2
(c) H_2 and He
(d) SO_x and NO_x

Answer: (d) SO_x and NO_x — They form H_2SO_4 and HNO_3.

85. In $4NO_2 + O_2 + 2H_2O \rightarrow X$, X is:

(a) $2HNO_3$
(b) $4HNO_3$
(c) $4HNO_2$
(d) $2H_2SO_4$

Answer: (b) $4HNO_3$ — Nitric acid formation.

86. Which toxic ion is made soluble by acid rain?

(a) Mg^{2+}
(b) K^+
(c) Ca^{2+}
(d) Al^{3+}

Answer: (d) Al^{3+} — It damages plant roots.

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87. Fish die in acidic lakes because Al^{3+}:

(a) Heats the water
(b) Removes their scales
(c) Clogs their gills
(d) Feeds algae

Answer: (c) Clogs their gills — Oxygen absorption falls and fish suffocate.

88. Acid rain lowers soyabean and wheat yields by reducing:

(a) Nitrogen fixation (b) Sunlight
(c) Soil temperature (d) Wind speed

Answer: (a) Nitrogen fixation — And nutrient intake.

89. Powdered limestone is added to acidic lakes to:

(a) Feed the fish (b) Neutralise the acid
(c) Add nitrogen (d) Kill weeds

Answer: (b) Neutralise the acid — $CaCO_3$ reacts with H_2SO_4.

32.8 CLIMATE CHANGE AND CROPS

90. The industry most vulnerable to shifting climates is:

(a) Cosmetics (b) Cement
(c) Agriculture (d) Steel

Answer: (c) Agriculture — Stated in section 32.8.

91. Heat stress in plants causes:

(a) Dehydration and poor photosynthesis
(b) Faster growth always
(c) More nitrogen fixation
(d) Higher yield

Answer: (a) Dehydration and poor photosynthesis — Growth and yield fall.

92. Changed rainfall patterns can cause:

(a) Acid-free soil
(b) Only higher yields
(c) More pollination
(d) Droughts and floods

Answer: (d) Droughts and floods — Both cause crop failure.

93. Cloud bursts take away the nutrient-rich top soil by:

(a) Leaching of Al^{3+} (b) Soil erosion
(c) Nitrogen fixation (d) Salination

Answer: (b) Soil erosion — Unpredictable rains erode soil.

94. "Sub1 rice" can survive under water for:

(a) Two years (b) Two days
(c) Two months (d) Two weeks

Answer: (d) Two weeks — A flood-proof variety.

95. An example of a climate-resilient crop is:

(a) Heat-tolerant wheat (b) BT-cotton
(c) Golden Rice (d) Margarine

Answer: (a) Heat-tolerant wheat — Also drought-resistant corn.

32.9 GENETIC ENGINEERING

96. Genetic engineering changes an organism's:

(a) Fertilizer (b) Soil
(c) Water supply (d) DNA

Answer: (d) DNA — By adding, removing or modifying genes.

97. GM stands for:

(a) Generally mixed (b) Genetically modified
(c) Gene mapped (d) Growth maximised

Answer: (b) Genetically modified — GM crops have desired traits.

98. A gene can be inserted into a plant genome using a:

(a) Gene gun (b) Rotary kiln
(c) Membrane cell (d) Fractionating column

Answer: (a) Gene gun — Or a carrier bacterium.

99. The correct order of steps in genetic engineering is:

(a) Expression, insertion, isolation, identification
(b) Insertion, isolation, expression, identification
(c) Identification, isolation, insertion, expression
(d) Isolation, expression, identification, insertion

Answer: (c) Identification, isolation, insertion, expression — As in Figure 32.4.

100. BT-cotton:

(a) Needs no water
(b) Produces its own pesticide
(c) Contains beta-carotene
(d) Grows under water

Answer: (b) Produces its own pesticide — It reduces chemical sprays.

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101. Golden Rice is engineered with:

(a) Chlorine (b) Caffeine
(c) Penicillin (d) Beta-carotene

Answer: (d) Beta-carotene — To prevent blindness.

102. The spread of GM genes to wild plants is called:

(a) Outcrossing (b) Leaching
(c) Mercerizing (d) Fortification

Answer: (a) Outcrossing — A concern about GM crops.

103. "Superweeds" are weeds that are resistant to:

(a) Sunlight (b) Rain
(c) Herbicides (d) Fertilizers

Answer: (c) Herbicides — Resistance genes jump by cross-pollination.

32.10–32.11 RAW MATERIALS AND ECONOMY

104. Air provides which gas for fertilizers?

(a) Carbon dioxide (b) Argon
(c) Nitrogen (d) Helium

Answer: (c) Nitrogen — Oxygen from air is used in steel making.

105. Fossil fuels provide hydrocarbons for about what share of organic chemicals?

(a) 10% (b) 90%
(c) 30% (d) 50%

Answer: (b) 90% — Stated in section 32.10.

106. The main component of natural gas is:

(a) Ethene (b) Ethane
(c) Butane (d) Methane

Answer: (d) Methane — CH_4; used for urea and petrochemicals.

107. One of the world's largest rock salt deposits is at:

(a) Khewra (b) Muslim Bagh
(c) Sui (d) Thar

Answer: (a) Khewra — Source of NaCl.

108. Limestone is the raw material for the:

(a) Textile industry (b) Cement industry
(c) Cosmetics industry (d) Urea industry

Answer: (b) Cement industry — It supplies $CaCO_3$.

109. Which mineral is used as a soil conditioner?

(a) Sulphur (b) Chromite
(c) Rock salt (d) Gypsum

Answer: (d) Gypsum — Also used in cement and plaster.

110. Chromite is found in:

(a) Lahore (b) Khewra
(c) Muslim Bagh (d) Karachi

Answer: (c) Muslim Bagh — Used for stainless steel and pigments.

111. The main raw material for making urea in Pakistan is:

(a) Natural gas (b) Rock salt
(c) Limestone (d) Chromite

Answer: (a) Natural gas — Local supply lowers the cost.

112. Textiles account for what share of Pakistan's exports?

(a) About 25% (b) About 5%
(c) About 15% (d) Over 50%

Answer: (d) Over 50% — Stated in section 32.11.

113. Caustic soda is used in the textile industry for:

(a) Dyeing fabric
(b) Bleaching fabric
(c) Mercerizing cotton
(d) Spinning yarn

Answer: (c) Mercerizing cotton — Hydrogen peroxide is used for bleaching.

114. The chemical used for bleaching fabrics is:

(a) Hydrogen peroxide (b) Urea
(c) Soda ash (d) Gypsum

Answer: (a) Hydrogen peroxide — Named in section 32.11.

32.12 INDUSTRIAL CHEMICAL PROCESSES

115. In addition polymerisation, monomers join:

(a) With loss of HCl
(b) With loss of water
(c) Without losing any atoms
(d) With loss of CO_2

Answer: (c) Without losing any atoms — Example: ethene to polyethene.

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116. Polyesters are made by:

(a) Addition polymerisation
(b) Condensation polymerisation
(c) Cracking
(d) Reforming

Answer: (b) Condensation polymerisation — A small molecule, usually water, is released.

117. The sole producer of PVC in Pakistan is:

(a) Lucky Core Industries
(b) Sitara Chemical Industries
(c) Attock Refinery
(d) Engro Polymer and Chemicals

Answer: (d) Engro Polymer and Chemicals — Stated in section 32.12.

118. Crude oil is separated into fractions by:

(a) Fractional distillation
(b) Electrolysis
(c) Filtration
(d) Crystallisation

Answer: (a) Fractional distillation — Based on boiling points.

119. Catalytic reforming gives gasoline of:

(a) Low octane number
(b) High octane number
(c) High sulphur content
(d) Low volatility

Answer: (b) High octane number — Straight chains become branched.

120. The catalyst in the Haber-Bosch process is:

(a) Nickel
(b) Platinum
(c) Iron
(d) V_2O_5

Answer: (c) Iron — Iron metal crystals.

121. The share of ammonia used directly for fertilizers is:

(a) 70%
(b) 13%
(c) 30%
(d) 50%

Answer: (a) 70% — The process gives 13% of the earth's nitrogen fixation.

122. Sulphuric acid is manufactured by the:

(a) Chlor-alkali process
(b) Solvay process
(c) Haber-Bosch process
(d) Contact process

Answer: (d) Contact process — SO_2 is oxidised to SO_3.

123. The catalyst in the Contact process is:

(a) Fe
(b) V_2O_5
(c) Ni
(d) Zeolite

Answer: (b) V_2O_5 — Platinum can also be used.

124. The "King of Chemicals" is:

(a) Ammonia (b) Nitric acid
(c) Sulphuric acid (d) Caustic soda

Answer: (c) Sulphuric acid — It is the most utilised chemical in the world.

125. The Solvay process produces:

(a) Ammonia (b) Sodium hydroxide
(c) Sulphuric acid (d) Sodium carbonate

Answer: (d) Sodium carbonate — From brine and limestone.

126. The by-product of the Solvay process is:

(a) $CaCl_2$
(b) NaCl
(c) NH_4NO_3
(d) H_2SO_4

Answer: (a) $CaCl_2$ — Ammonia is regenerated.

127. The precipitate formed in the Solvay process is:

(a) $CaSO_4$ (b) NaOH
(c) $NaHCO_3$ (d) NaCl

Answer: (c) $NaHCO_3$ — Heating converts it to Na_2CO_3.

128. The chlor-alkali process is carried out by:

(a) Burning sulphur (b) Heating limestone
(c) Cracking crude oil (d) Electrolysis of brine

Answer: (d) Electrolysis of brine — Modern plants use membrane cells.

129. Which is NOT a product of the chlor-alkali process?

(a) Chlorine (b) Sodium carbonate
(c) Sodium hydroxide (d) Hydrogen

Answer: (b) Sodium carbonate — Soda ash comes from the Solvay process.

32.13–32.14 APPLICATIONS AND SAFETY

130. Surfactants are used in cosmetics for making:

(a) Margarine (b) Perfume bottles
(c) Cement (d) Soaps

Answer: (d) Soaps — Emulsifiers keep creams smooth.

131. Preservatives are added to cosmetics to prevent:

(a) Colour (b) Bacterial growth
(c) Foaming (d) Fragrance

Answer: (b) Bacterial growth — Stated in section 32.13.

132. Cement is made by heating limestone and clay to about:

(a) 500 °C
(b) 150 °C
(c) 1500 °C
(d) 5000 °C

Answer: (c) 1500 °C — In a rotary kiln.

133. Cement clinker is ground with:

(a) Gypsum
(b) Rock salt
(c) Chromite
(d) Sulphur

Answer: (a) Gypsum — The final step of manufacture.

134. Cement becomes warm on adding water because hydration is:

(a) Photochemical
(b) Endothermic
(c) Reversible
(d) Exothermic

Answer: (d) Exothermic — Heat is released.

135. A common food preservative is:

(a) Sodium carbonate
(b) Sodium benzoate
(c) Sodium hydroxide
(d) Sodium chloride brine

Answer: (b) Sodium benzoate — Named in section 32.13.

136. Edible oils are turned into margarine by:

(a) Fermentation
(b) Fortification
(c) Hydrogenation
(d) Cracking

Answer: (c) Hydrogenation — Oils become solid fats.

137. Baking soda acts in baking as a:

(a) Leavening agent
(b) Preservative
(c) Emulsifier
(d) Surfactant

Answer: (a) Leavening agent — It is sodium bicarbonate.

138. Adding iodine to salt is an example of:

(a) Hydrogenation
(b) Fortification
(c) Leavening
(d) Mercerizing

Answer: (b) Fortification — Vitamin A is added to oil in the same way.

139. PPE stands for:

(a) Plant process equipment
(b) Public protection engineering
(c) Personal protective equipment
(d) Personal pollution exposure

Answer: (c) Personal protective equipment — Gloves, goggles, helmets, coats, shoes.

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140. Flammable, toxic and corrosive chemicals should be stored:

(a) In suitable containers in separate areas
(b) Together in one tank
(c) In open trays
(d) Near heat sources

Answer: (a) In suitable containers in separate areas — This prevents leaks and dangerous reactions.

141. Regular inspection of machines and pipelines helps to:

(a) Avoid training
(b) Increase fumes
(c) Remove labels
(d) Detect faults early

Answer: (d) Detect faults early — It reduces the risk of accidents.

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